

Task-visible Inputs



S: Task specification

instruction.md, task description



P: Executable PyTorch repo

reference implementation for development



Released checkpoint

model weights



Current JAX repository

$J_{(t-1)}$
candidate under development



Previous evidence $E_{(t-1)}$

validated and unresolved properties so far

Available during development



Diagnose–Migrate–Verify Loop



Diagnoser

localize discrepancies

- analyze evidence
- propose repair objectives D_t
- prioritize next migration



Migrator

update candidate repository

- edit JAX/Flax code
- revise checkpoint conversion
- modify training / serving



Verifier

produce validated evidence

- exercise updated repository
- record established vs unresolved properties
- return E_t

$$J_{(t-1)} \rightarrow D_t \rightarrow J_t \rightarrow E_t \rightarrow D_{(t+1)}$$

Repository state and validated diagnostic state are maintained separately.



Reference-Grounded Diagnostics $G(J; S, P)$



A. Differential against P

- output-contract verification
- compare runtime structure / fields
- anchor checks to behavior outside the candidate



B. Candidate-side checks

- module import
- forbidden dependency / no-network
- parameter tree
- gradient liveness
- JAX tracing
- training entry



C. Diagnostic outcomes



established



violated



not established

Probe failure is distinguished from candidate failure.